Hasil Simulasi Algoritma Goertzel di MATLAB

Kode MATLAB (1)

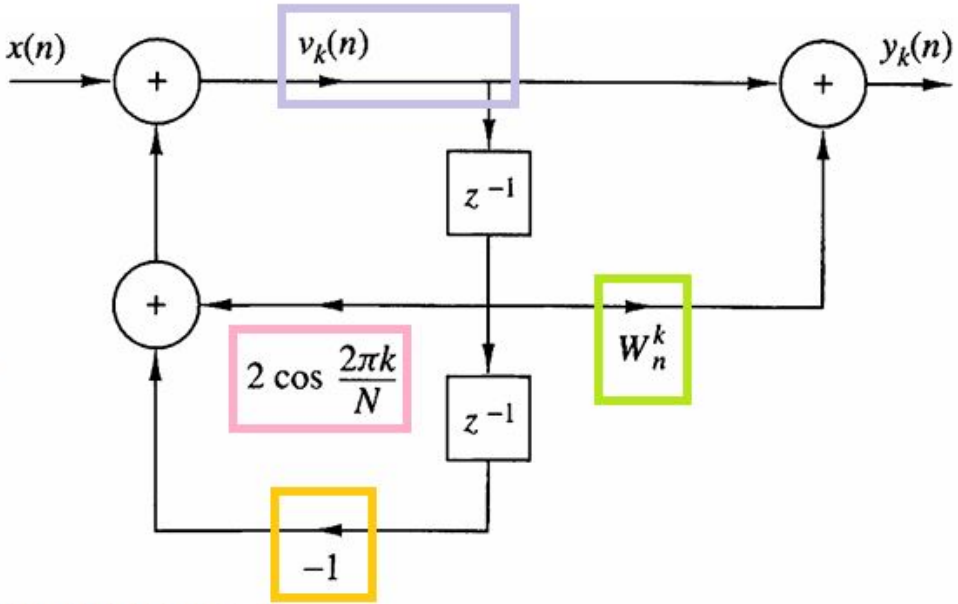
```
function [yn, Ak, filter_output] = goertzel4(x, k)

    N = length(x);

    x = [x 0];
    vk = zeros(1, N+3);
    filter_output = zeros(1, N);

    for n = 1:N+1
        vk(n+2) = 2*cos(2*pi*k/N)*vk(n+1) - vk(n) + x(n);
        if n <= N
            filter_output(n) = vk(n+2);
        end
    end

    yn = vk(N+3) - exp(-2*pi*1i*k/N)*vk(N+2);
    Xk = vk(N+3)*vk(N+3) + vk(N+2)*vk(N+2) - 2*cos(2*pi*k/N)*vk(N+3)*vk(N+2);
    Ak = sqrt(Xk)/N;
end
```



Kode MATLAB (2)

new

```
for n = 1:N+1
    vk(n+2) = 2*cos(2*pi*k/N)*vk(n+1) - vk(n) + x(n);
    if n <= N
        filter_output(n) = vk(n+2);
    end
end
```

old

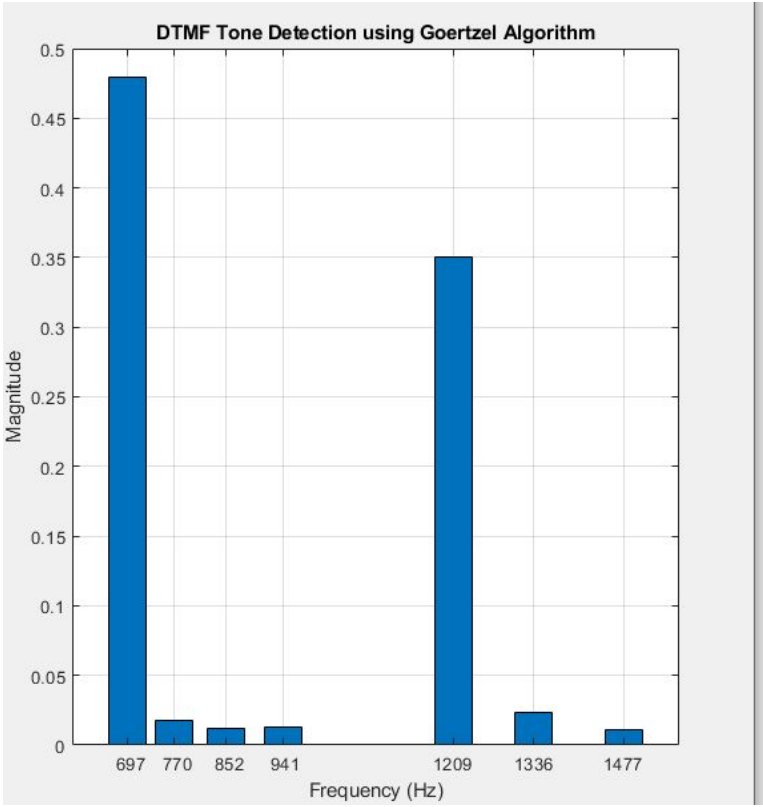
```
for m = 3:length(Y)
    Y(m) = (2 * cos(2*pi*k) * Y(m-2)) - Y(m-1) + X(m);
end
```

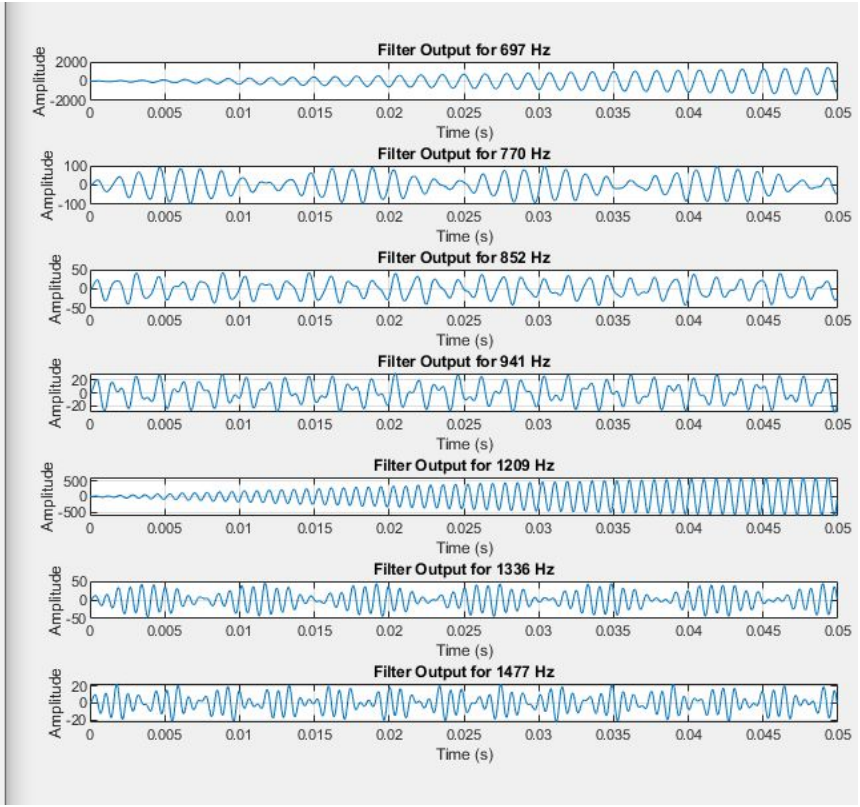
k_new = round(f * N/ Fs)
—-> pembagian oleh N dilakukan 2x (saat deklarasi k dan saat assignment vk(n+2)

k_old = (f * N/Fs)
—> pembagian oleh N hanya dilakukan 1x (saat deklarasi k)

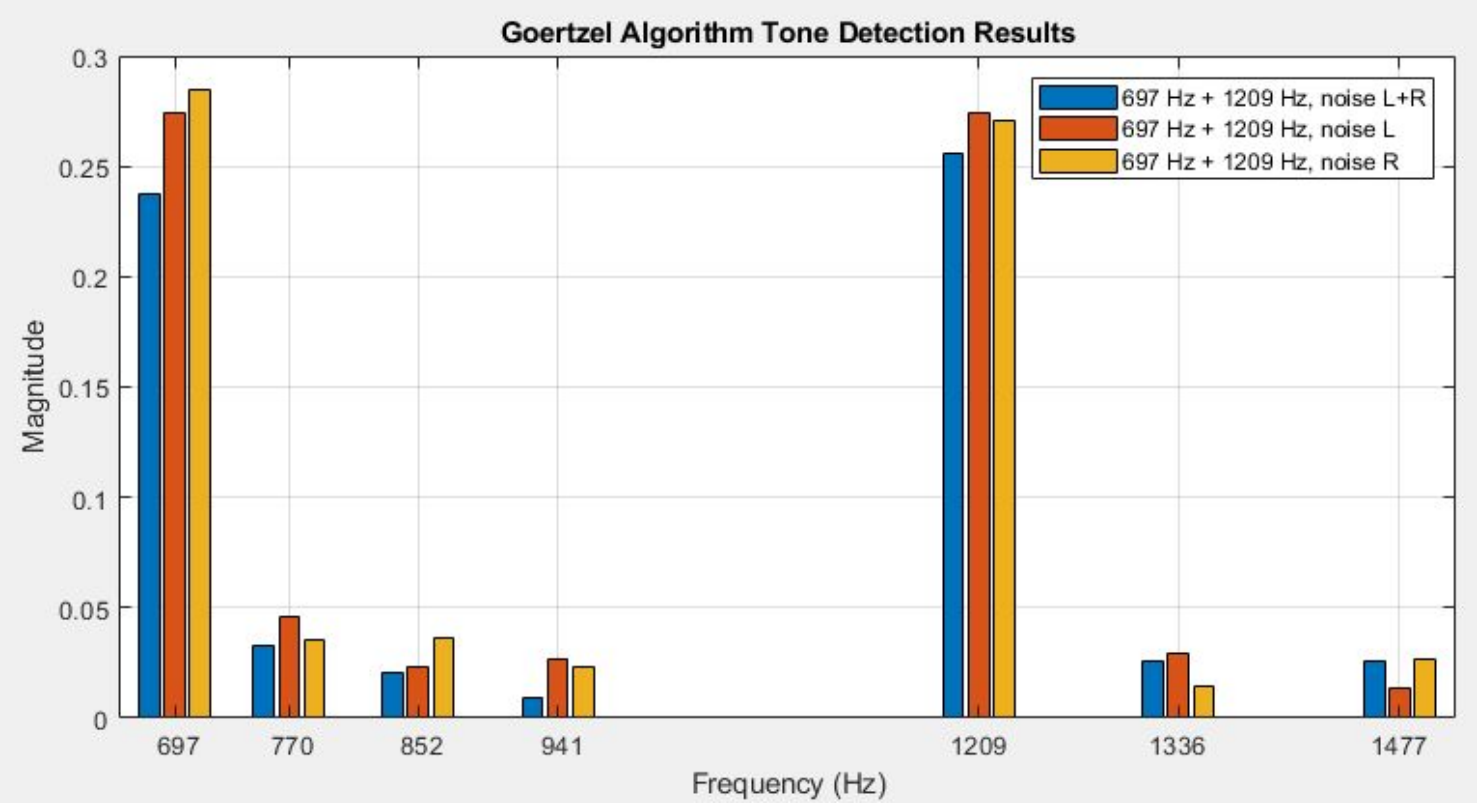
Catatan: meski di kode lama digunakan Y(...), label yang lebih tepat adalah vk(...)

vk(n) dan deteksi DTMF tanpa noise

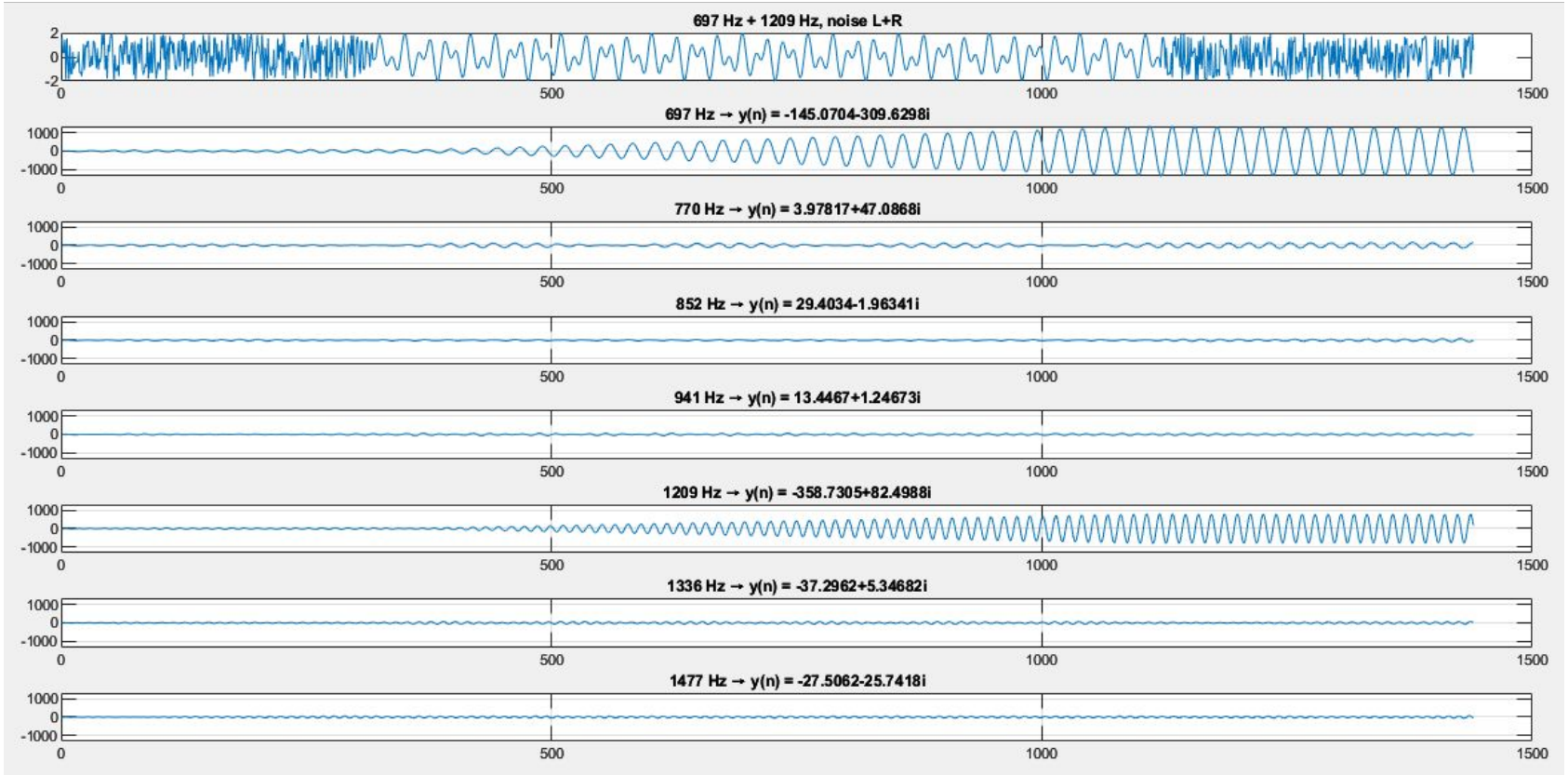




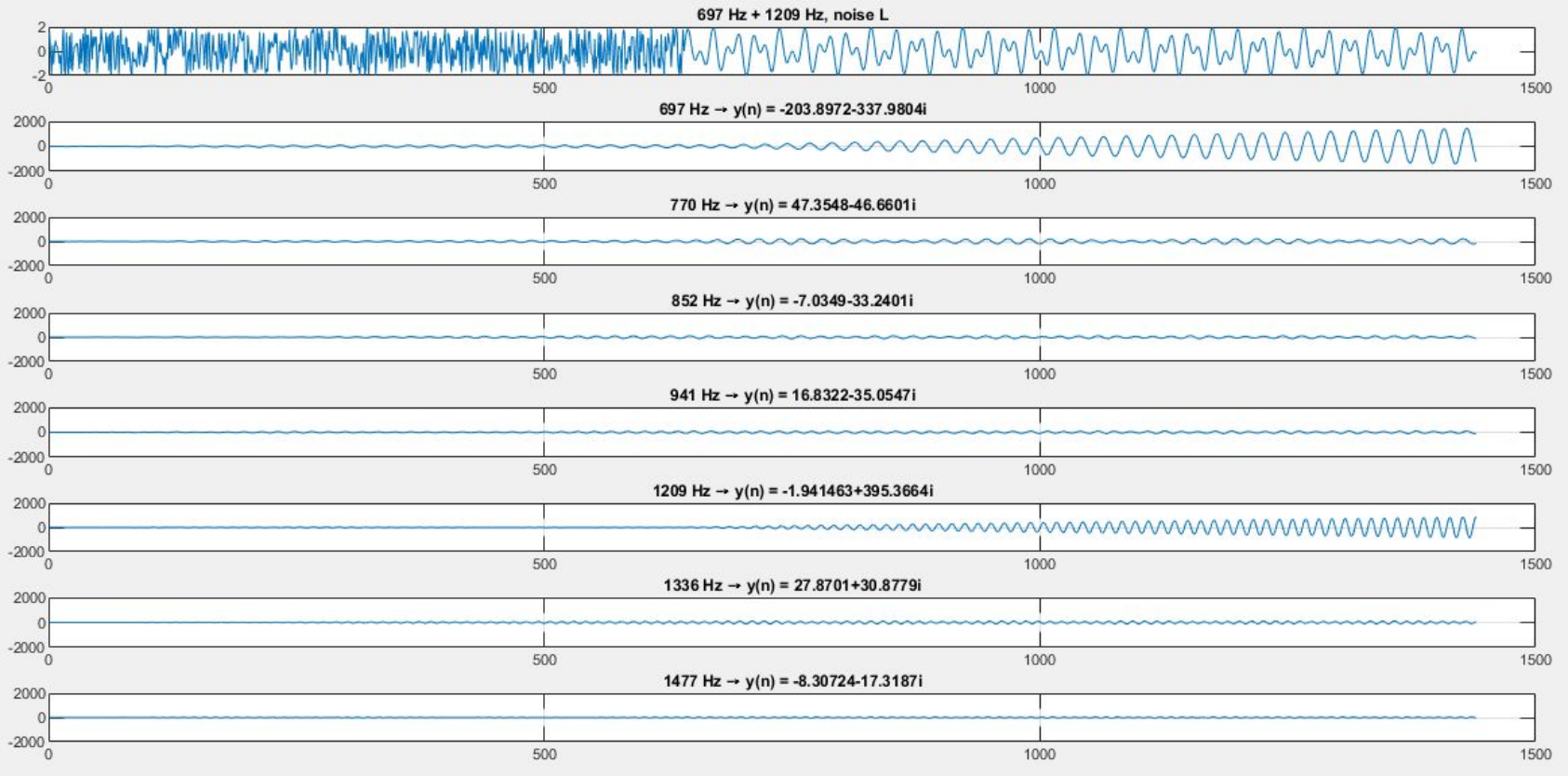
vk(n) dan deteksi DTMF dengan noise, 697 Hz + 1209 Hz



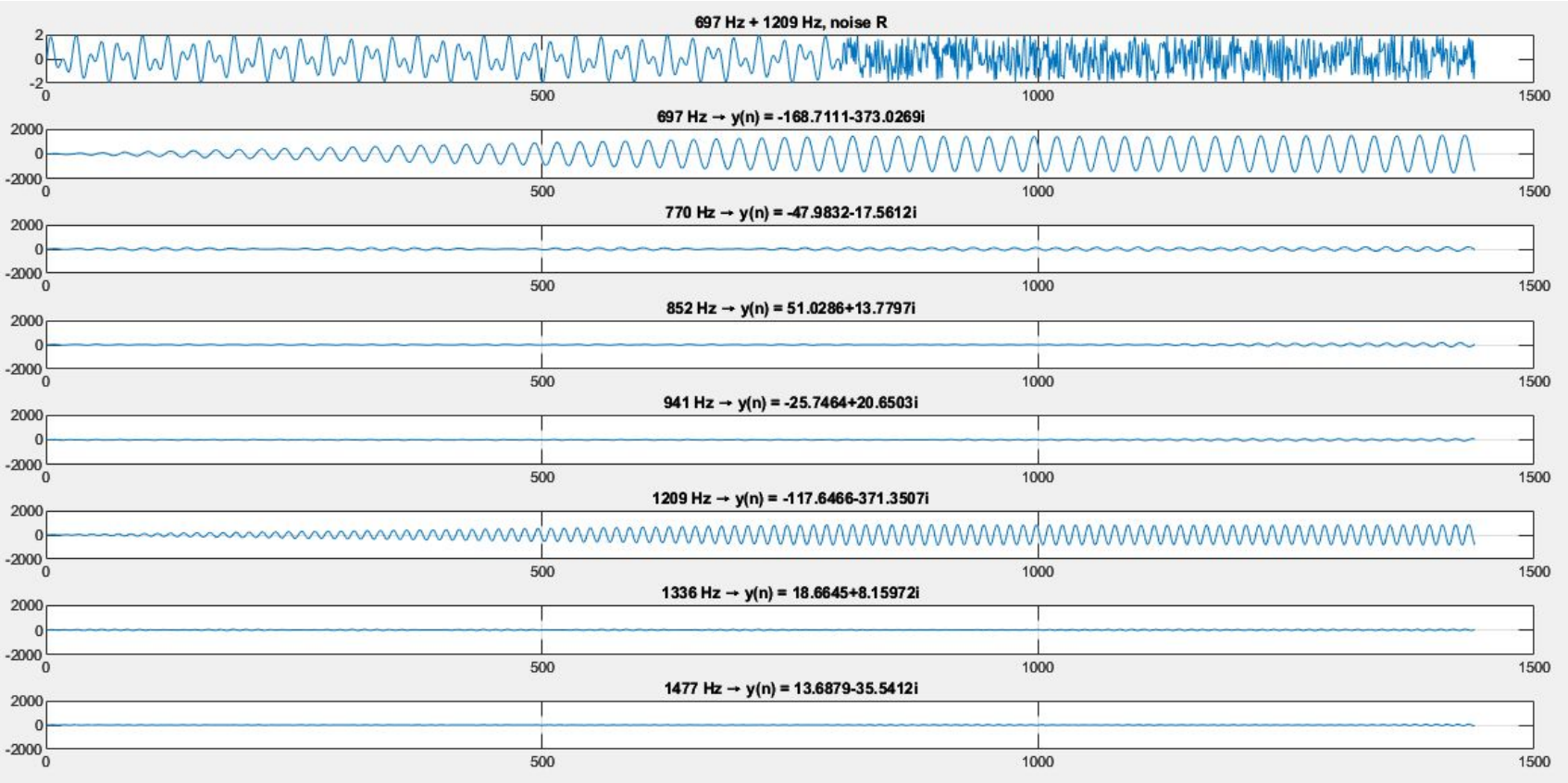
vk(n) dan y(n) untuk input dengan noise (1)



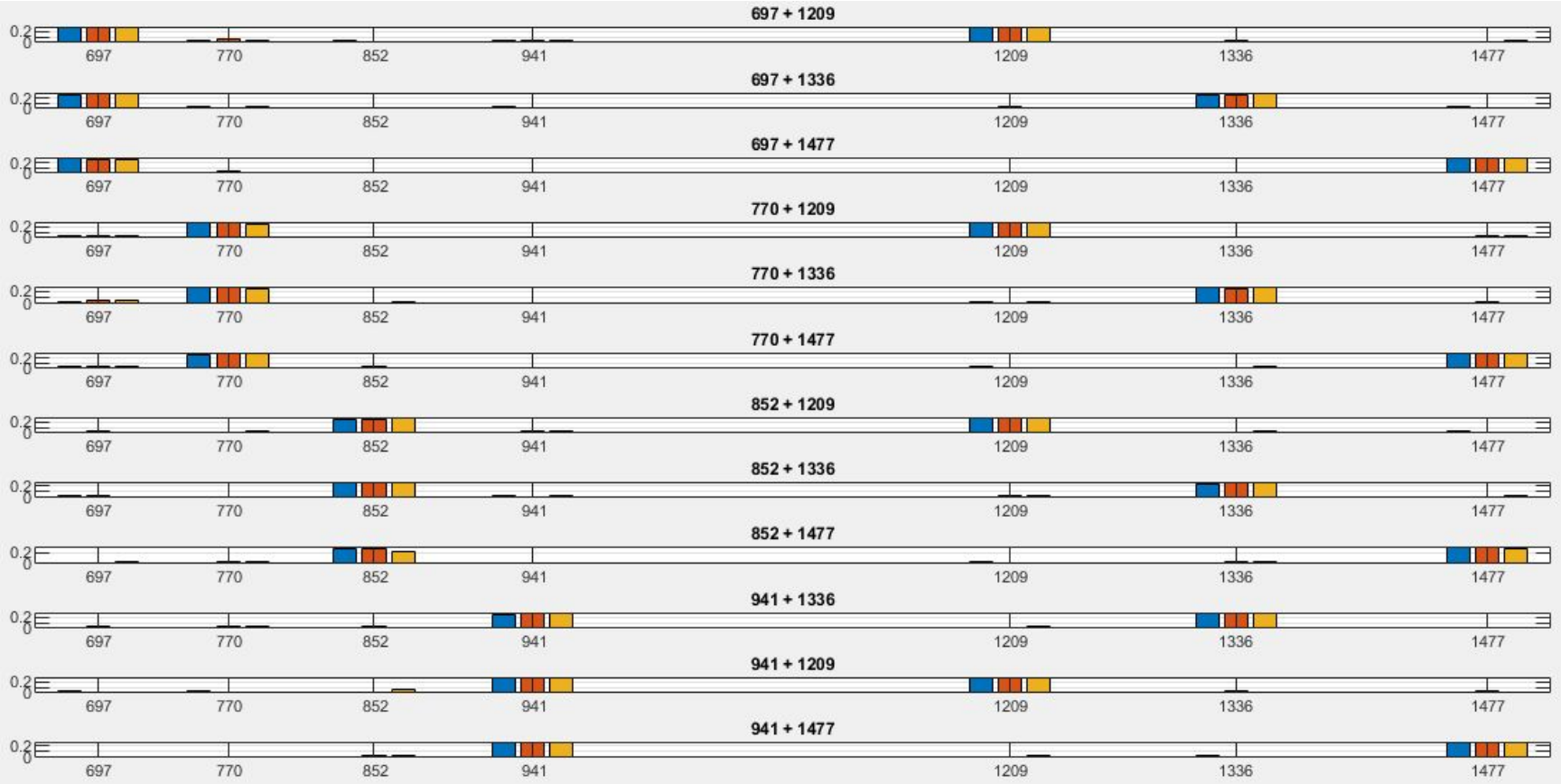
vk(n) dan y(n) untuk input dengan noise (2)



vk(n) dan y(n) untuk input dengan noise (3)

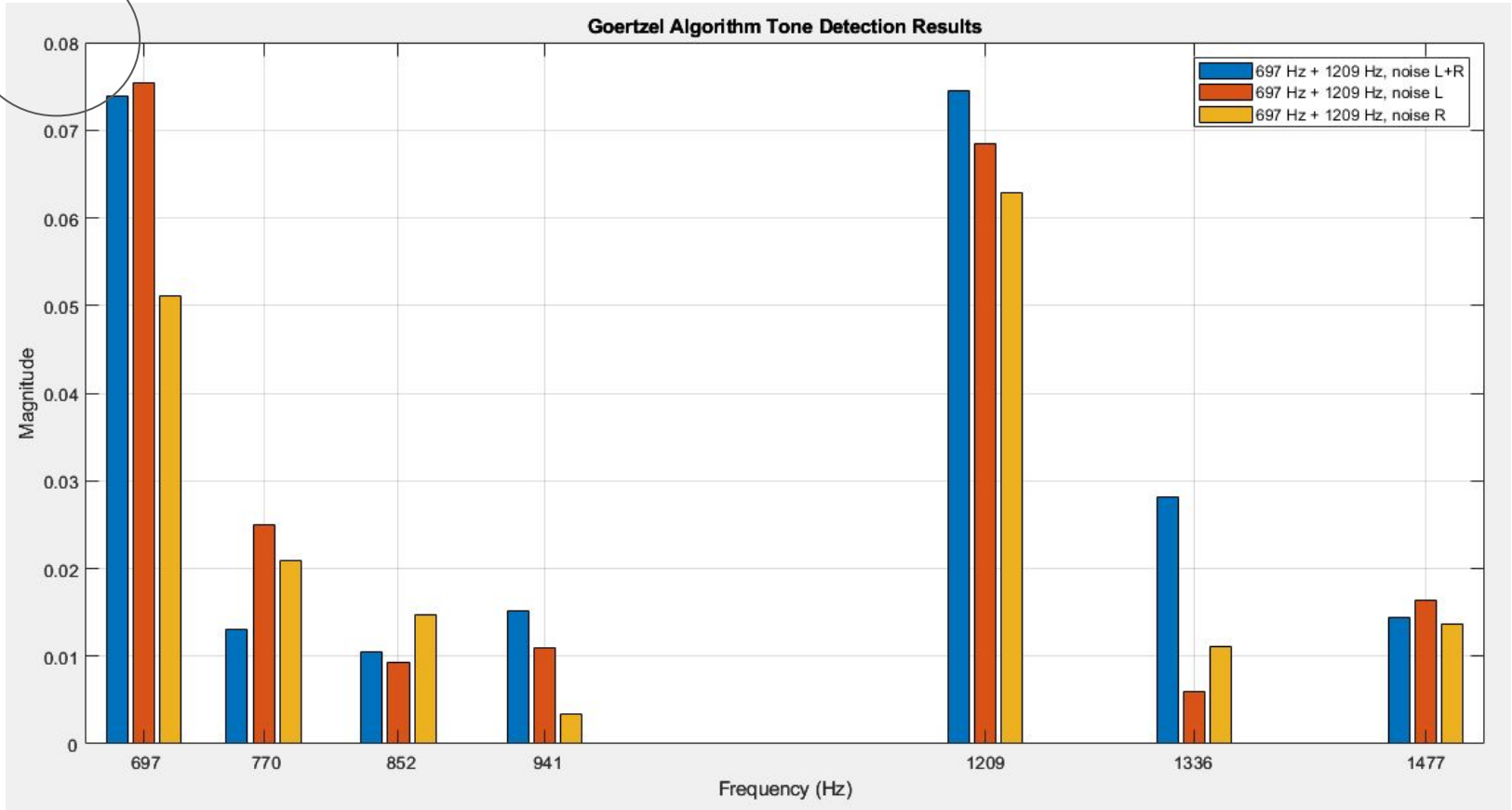


Deteksi DTMF untuk input dengan noise, 12 nada

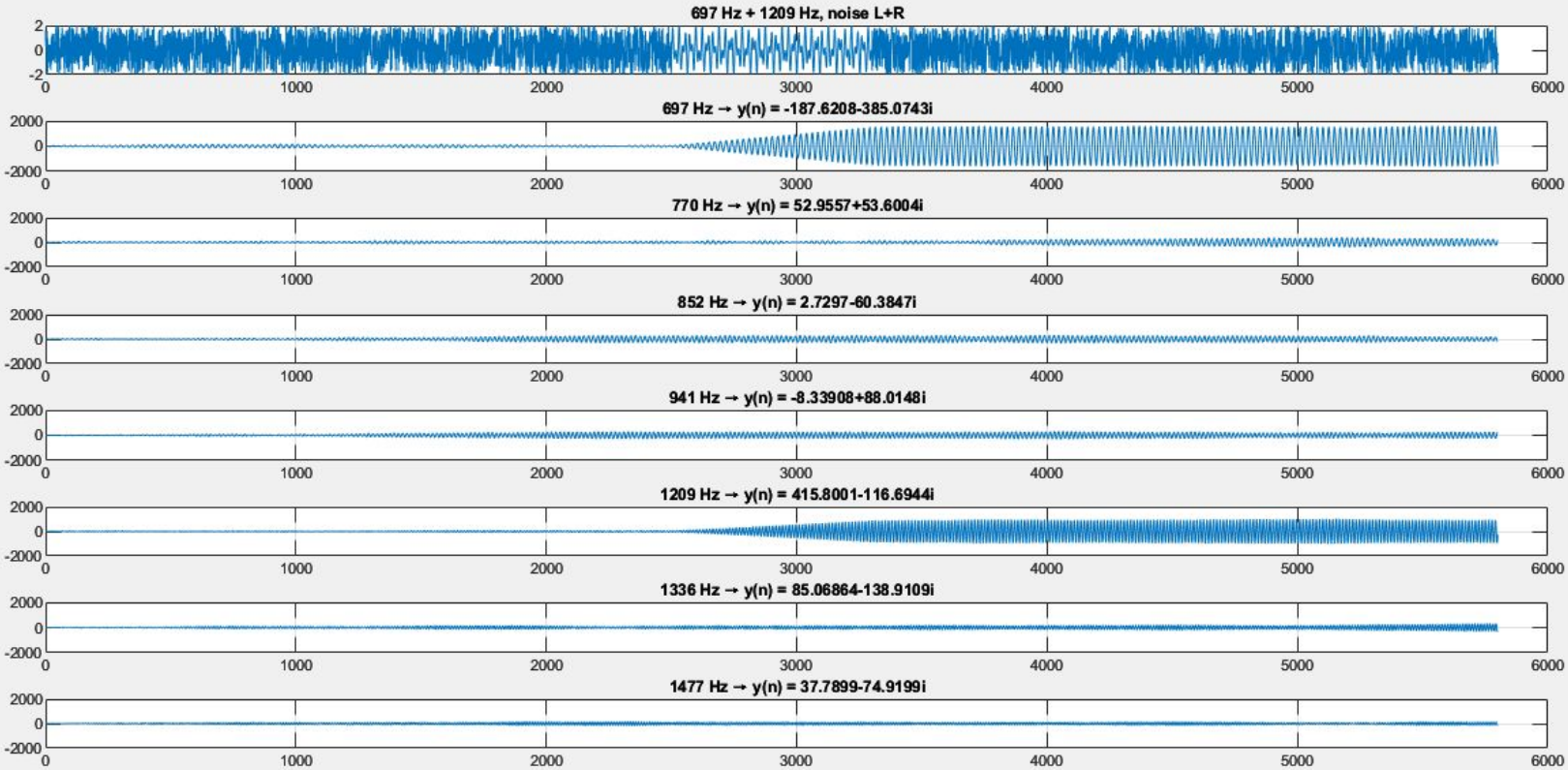


4x smaller magnitude

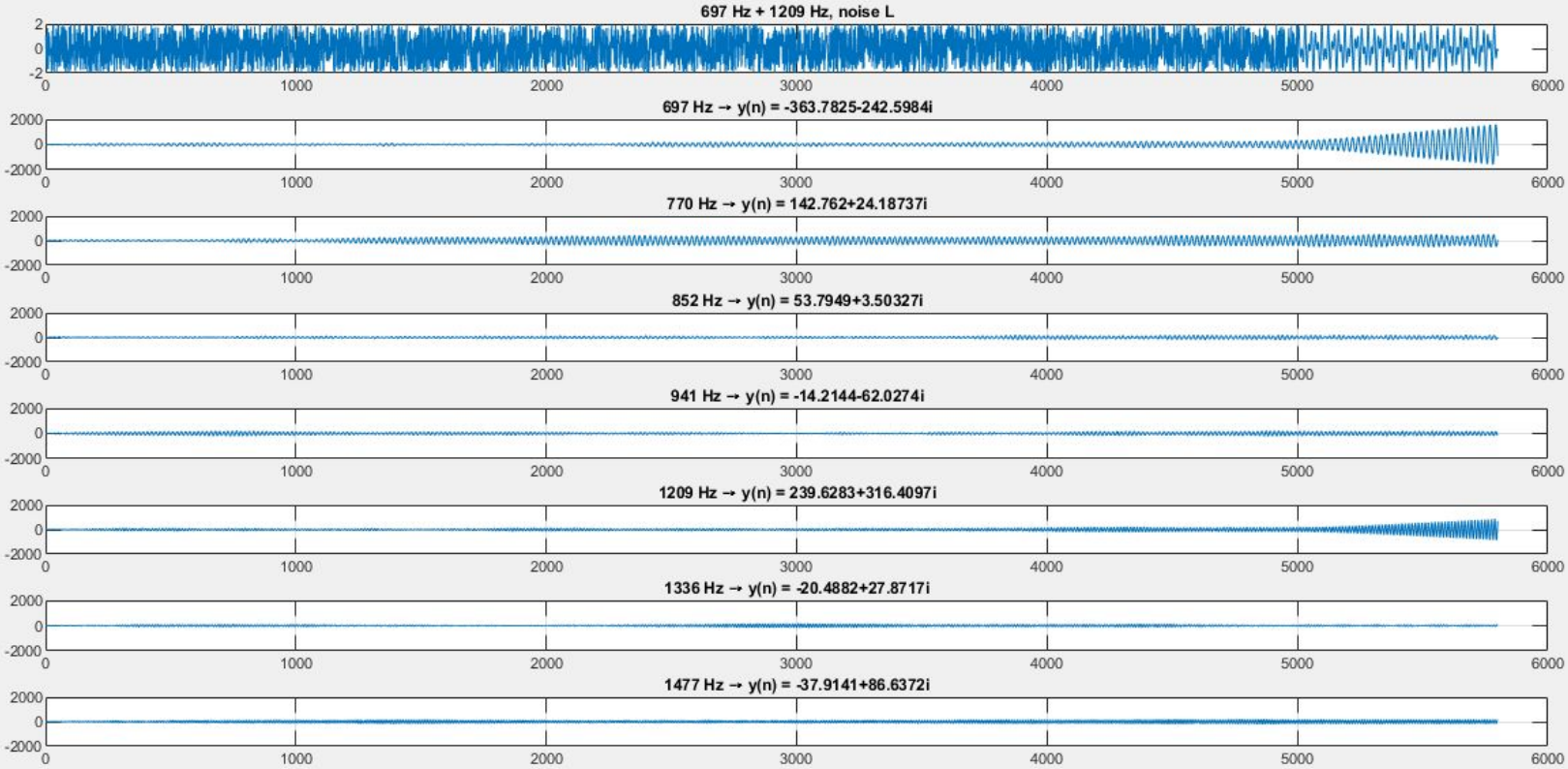
Longer inputs, 5000 noise samples (1)



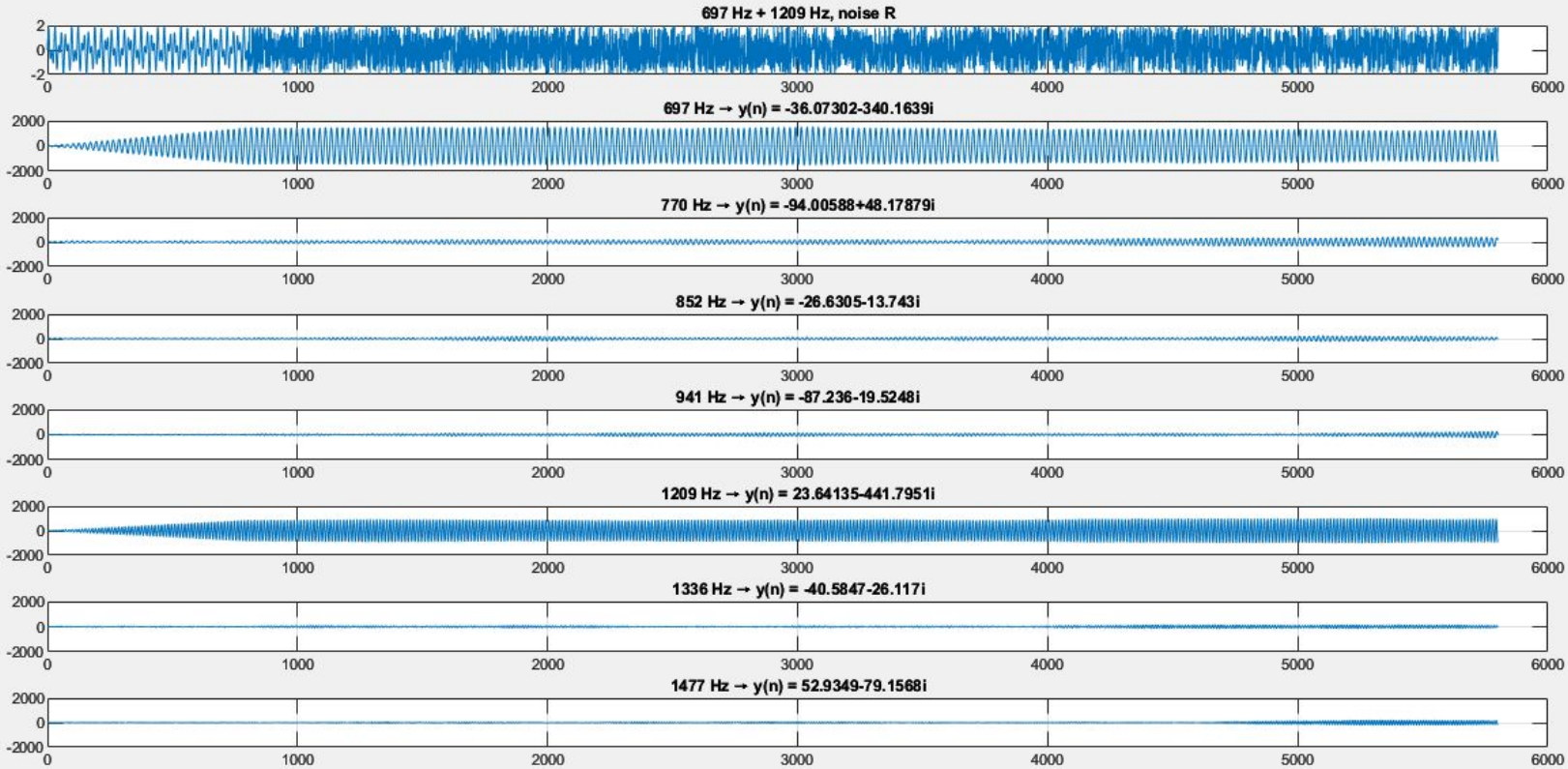
Longer inputs, 5000 noise samples (2)



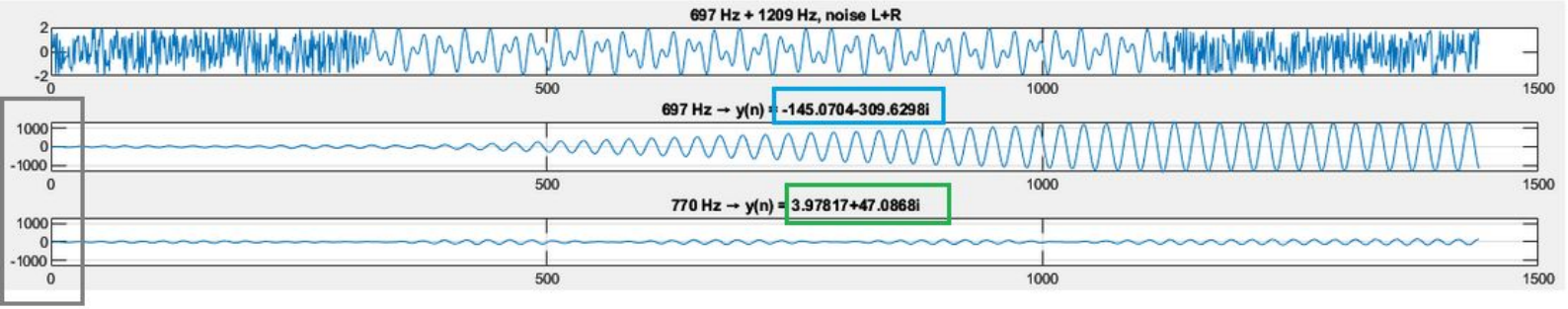
Longer inputs, 5000 noise samples (3)

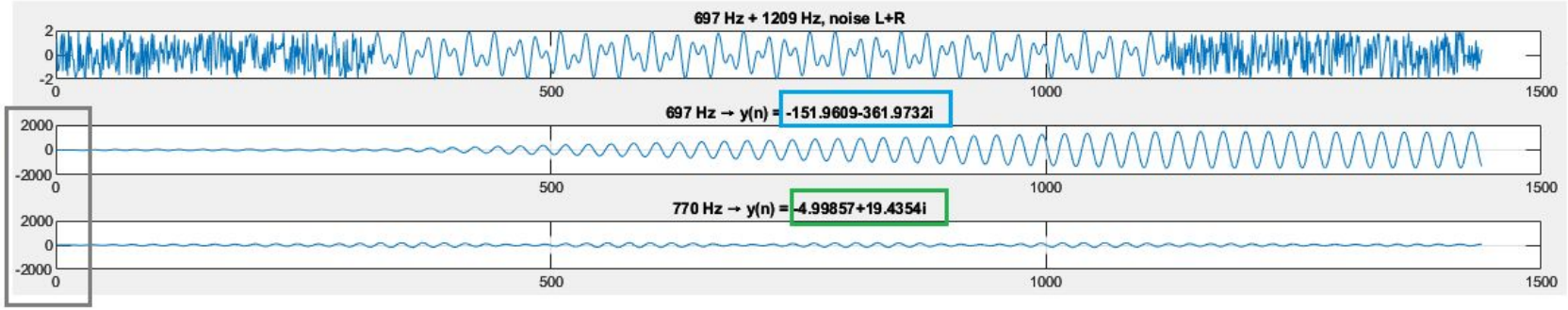


Longer inputs, 5000 noise samples (4)



Double vs Single Precision



double precision
(MATLAB default)

single precision
using single()